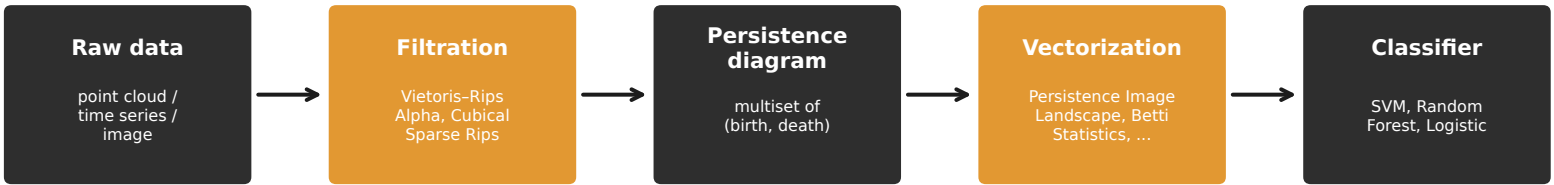The three-stage persistent homology classification pipeline